

The image features the Flutter logo, which consists of a blue parallelogram and a green parallelogram, positioned in the upper left quadrant. The background is a dark blue gradient with abstract, light gray geometric patterns, including a circular circuit-like design on the left and a series of parallel lines on the right. The word "Flutter" is written in a white, sans-serif font in the center-right area.

Flutter



Creating a Flutter Project

Basic commands

- flutter help
 - Help command for checking what commands you can use
- flutter create --platforms=android example --empty
 - command for create a platform specific flutter app and clean.
- flutter pub add <package_name>
 - add a package/dependency
- flutter pub get
 - fetch all dependencies listed in pubspec.yaml
- flutter clean
 - clean the project dependencies

pubspec.yaml

- this is where you manage your project configuration and project dependencies(packages)



`main()`

- The main function is the entry point to any Dart program, and a Flutter app is just a Dart program.

`runApp()`

- The `runApp` method is part of the Flutter SDK, and it takes a widget as an argument.



MaterialApp

- MaterialApp is the root widget for apps using Material Design.
- Provides navigation, themes, routes, and app configuration.

Scaffold

- Scaffold provides the basic structure of a Flutter screen.
- Includes AppBar, Body, Drawer, FloatingActionButton, and BottomNavigationBar.

AppBar

- AppBar creates the top navigation bar of a screen.
- Usually contains title, icons, and actions.

SafeArea

- SafeArea prevents UI content from being hidden by system areas.
- Example: phone notch, status bar, or navigation bar.



Text & Display widgets

Text

- Text widget is used to display a string of text on the screen.
- Supports styling like font size, color, weight, alignment, and font family.

RichText

- RichText widget is used to display text with multiple styles inside one text block.
- Example: different colors or bold parts in a sentence.

Icon

- Icon widget is used to display icons from Flutter's built-in icon library.
- Commonly used for buttons, menus, and indicators.

Image

- Image widget is used to display images.
- Can load images from assets, network URLs, files, or memory.

Placeholder

- Placeholder widget displays a temporary box while developing a UI.
- Useful when planning layouts before adding real widgets.



Layout widgets

Container

- Container widget is a box that can contain other widgets.
- Used for adding padding, margin, size, color, borders, and decoration.

SizeBox

- SizeBox widget creates a box with a fixed width and height.
- Commonly used for spacing between widgets.

Padding

- Padding widget adds empty space around its child widget.
- Used to control the distance between content and its boundary.

Center

- Center widget positions its child in the middle of available space.

Align

- Align widget positions a child using alignment rules.
- Example: top-left, bottom-right, center.



Row

- Row widget arranges children horizontally.
- Example: placing icons or buttons beside each other.

Column

- Column widget arranges children vertically.
- One of the most commonly used layout widgets.

Stack

- Stack widget places widgets on top of each other.
- Used for overlapping UI elements.
- Example: text over an image.

Expanded

- Expanded widget makes a child fill available space inside Row or Column.
- Uses flexible layout behavior.

Flexible

- Flexible widget allows a child to resize within Row or Column.
- Gives more control than Expanded.

Wrap

- Wrap widget automatically moves children to the next line when space runs out.
- Useful for tags, chips, and buttons.



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Input Widgets

TextField

- TextField allows users to enter text.
- Commonly used for search bars, login forms, and inputs.

Button Widgets

ElevatedButton

- ElevatedButton creates a raised button with a background color.
- Commonly used for primary actions.

TextButton

- TextButton creates a simple button without a background.
- Used for secondary actions.

OutlinedButton

- OutlinedButton creates a button with a border.
- Used when you want a less emphasized action.

IconButton

- IconButton creates a clickable icon button.
- Commonly used in AppBars.



Lists & Scrolling Widgets

ListView

- ListView displays a scrollable list of widgets.
- Commonly used for feeds, menus, and dynamic data.

GridView

- GridView displays items in a grid layout.
- Example: photo galleries or product lists.

SingleChildScrollView

- SingleChildScrollView allows a single widget to become scrollable.

ListTile

- ListTile provides a standard list item layout.
- Includes title, subtitle, leading icon, and trailing widget.



reference

<https://api.flutter.dev/flutter/material/>

<https://pub.dev/>